SWE 4304 : Software Project Lab I

Islamic University of Technology

Dormatrix

Project Proposal

Team-1

Team Members

Ayman Binta Altaf Nondiny - 230042141

Khadiza Sultana - 230042135

Procheta Silvie - 230042114

Sayma Tasnim - 230042139

Introduction

- Digitalizing the management of dormitory
- Streamlining hall operations in a centralized platform
- Enhancing experience for both students and authorities
- Efficient monitoring of facilities and resources.

Motivation

Existing Problems in the IUT Female Dormitory:

- Unscheduled worker visits
- Delayed complaint resolution process
- No unified facility-booking system
- Ineffective shared appliance system
- Disorganized departmental store credit tracking

Key Features

- Role-Based Access Control
- Hall Office Management Tools
- Departmental Store Module
- Service & Maintenance Tracker
- Appliance & Utility Manager
- Facility Booking Manager

Multiple User Views in Dormatrix

- Students
- Attendants
- Maintenance Staff
- Store In-Charge
- Hall Office
- System Administrator

Resident View

- **Student Dashboard** provides access to:
  - Room Dashboard
  - Compaint Submission
  - Departmental Store
  - Lost & Found
  - Emergency Health Assistance
  - Events & Notices

Attendants & Staff

- **For Attendants:**
  - Centralized hall overview
  - Easy complaint tracking
  - Store dues tracking
- **For Maintenance Staff:**
  - Availability Check
  - Status update options
  - Priority based task queue

Admin & Others

- **For System Admin:**
  - User account management
  - Permission control
  - Backup & logging system
- **For Store In-Charge:**
  - Account purchases
  - Inventory tracking
- **For Hall Office:**
  - Resident info management
  - Automated report generation

Technical Mechanisms I

Password Authentication

- Manual hashing
- Role-based access control

Slot Memory Management

- Slots treated as memory frames
- Used in: Fridge slot allocation, Study-room seat booking

Technical Mechanisms II

Permission Mapping

- Each role has distinct access
- Controlled access across modules

Due Ledger

- Payment tracking
- Inventory tracking
- Dues tracking

Service Schedule Tracker

- Optimizes staff assignment
- Confirms task completion

Tools & Technologies

- **Language :** Java
- **File Based Storage :** text based file (.txt)
- **I/O:** Java FileReader / FileWriter / BufferedReader
- **Architecture:** Modular OOP
- **Version Control:** Github
- **SDLC Model:** Iterative + Incremental

Proposed Timeline

Thank You